

Macroeconomics I

Foundations of Dynamic General Equilibrium Models: Equilibrium, Welfare, & Uncertainty

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Introduction

- The Solow model matches some stylized facts about growth but relies on an exogenous, constant savings rate.
- We want to **microfound** the savings decision so it responds to prices and policy.
- We also want the model in **general equilibrium**: prices clear markets and respond to the environment.

In this lecture:

- How to define and solve for a competitive equilibrium.
- Arrow-Debreu vs. sequential market structure.
- What can we say about welfare (in the Pareto sense).
- How to include uncertainty in a dynamic model.

References

- Dirk Krueger Ch. 2 and 6.
- PhD Macrobook Ch. 5, 6 and 7.2.
- Acemoglu Ch. 5.
- Ljungqvist and Sargent. Ch. 8.

Ingredients of a (Macro)economic Model

- **Preferences:** Utility function.
- **Technology:** Production function.
- **Environment:** Information, market structure, goods, population.
- **Endowments:** Agents' initial resources.
- **Government:** Policy instruments and objective (when present).
- **Concept of equilibrium:** How prices are defined, or alternatively, how interactions between the agents occur.

Competitive Equilibrium:

Definition

Allocations and prices such that: (i) given prices, allocations solve each agent's problem; (ii) allocations satisfy the resource constraints (*market clearing*).

- The equilibrium concept specifies how prices coordinate decentralized decisions.

Solving the Model

Step by step:

1. Describe the *environment*.
2. Solve each agent's problem $\Rightarrow$ demand/supply as a function of prices.
3. Impose *market clearing* $\Rightarrow$ find the equilibrium prices.
4. Write the equilibrium: N equations, N unknowns.

Why bother?

- **Transparency:** clear separation between preferences/technology and equilibrium outcomes.
- **Micro** $\Rightarrow$ **Macro**: aggregate relationships respect individual constraints.
- Policy analysis: welfare effects are attributed to specific agents.

(Macro)economic Models

Macro models are...

D ynamic

S tochastic

G eneral

E quilibrium

Two-Agent Endowment Economy

- **Environment:** 2 consumers ($i = 1, 2$) of a single good living infinitely many periods.
- **Preferences:**

$$U(\{c_t\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t u(c_t^i) \quad i = 1, 2. \quad (1)$$

- **Endowments:** Deterministic sequences $\{e^i\}_{t=0}^{\infty}$, where:

$$e_t^1 = \begin{cases} \hat{e}, & \text{if } t \text{ is even.} \\ 0, & \text{if } t \text{ is odd.} \end{cases} \quad e_t^2 = \begin{cases} 0, & \text{if } t \text{ is even.} \\ \hat{e}, & \text{if } t \text{ is odd.} \end{cases} \quad (2)$$

and $\hat{e} > 0$.

- **“Technology”:** The endowment can be transformed into a final consumption good at no cost: $c_t = \hat{e}$

Digression: Infinite Horizon

Why $T = \infty$?

- (i) **Altruism:** utility from descendants' well-being implies
$$U(c_\tau) = u(c_\tau) + \beta U(c_{\tau+1}) = \sum_{t=\tau}^{\infty} \beta^{t-\tau} u(c_t).$$
- (ii) **Simplification:** for large T , behavior is similar to $T = \infty$; infinite-horizon models are stationary and easier to work with.

Convergence: we need $U = \sum_{t=0}^{\infty} \beta^t u(c_t)$ to be well-defined.

- For a constant sequence $c_t = \bar{c}$: convergence requires $\beta < 1$.
- For a growing sequence $c_t = c_0(1 + \gamma)^t$: convergence depends on γ , β , and the curvature of u .
- This imposes restrictions on parameters and functional forms.

Digression: Utility Function

Assumptions on u :

- Twice-differentiable, time-separable, defined over $c > 0$.
- Strictly increasing: $u'(c) > 0$.
- Strictly concave: $u''(c) < 0$.
- Inada conditions: $\lim_{c \rightarrow 0} u'(c) = \infty$ and $\lim_{c \rightarrow \infty} u'(c) = 0 \Rightarrow$ interior solution.

Other assumptions:

- **Exponential discounting:** $U^i = \sum_{t=0}^{\infty} \beta^t u(c_t^i)$, $\beta \in (0, 1)$. The discount between t and $t + 1$ is always β , regardless of t .
- **Homogeneous preferences:** all agents share the same u and β .
- **Infinite horizon:** $T = \infty$ (dynasties / stationarity).

Usual Utility Functions

- Usual utility functions used in macro models:

$$u(c) = \ln c \quad \text{Log}$$

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0 \quad \text{CRRA}$$

$$u(c) = \theta c, \quad \theta > 0 \quad \text{Linear}$$

$$u(c) = c - \theta \frac{c^2}{2} \quad \theta > 0 \quad \text{Quadratic}$$

$$u(c) = -\frac{\exp\{-\alpha c\}}{\alpha} \quad \alpha > 0 \quad \text{CARA}$$

- Note that $\ln c$ is a special case of CRRA when $\sigma = 1$.

Market Structure

- What is a decentralized equilibrium? Allocations supported by prices that clear all markets.
- Basically solving supply and demand in $N - 1$ markets (by Walras' law, the N -th market will be in equilibrium).

Two ways to represent a competitive equilibrium in a dynamic economy

1. **Arrow-Debreu:** All exchanges occur in period 0.
2. **Sequential Markets:** Markets open each period.

Arrow-Debreu Structure

- Agents “trade” in period 0 (or sign a contract with perfect commitment).
- In subsequent periods, they only deliver the quantities agreed upon in period 0.
- The price of the final consumption good is p_t in each t . We normalize $p_0 = 1$.
- Intuitively, a consumption good at t is a different commodity at $t - 1$ (thus has a different price).
- In *complete markets*, T periods are equivalent to having T different goods in a single period.
- The budget constraint for agent i in period 0: $\sum_{t=0}^{\infty} p_t c_t^i \leq \sum_{t=0}^{\infty} p_t e_t^i$.

Definition 1 (Arrow-Debreu competitive equilibrium)

A set of sequences $\{c_t^{i*}\}_{t=0}^{\infty}$ for $i = 1, 2$, and $\{p_t\}_{t=0}^{\infty}$ (with $p_0 = 1$) such that:

1. For each i , $\{c_t^{i*}\}_{t=0}^{\infty}$ solves

$$\max_{\{c_t^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t^i) \quad \text{subject to} \quad \sum_{t=0}^{\infty} p_t c_t^i = \sum_{t=0}^{\infty} p_t e_t^i.$$

2. For each t , $c_t^{1*} + c_t^{2*} = e_t^1 + e_t^2 = \hat{e}$.

- Condition 1: individual optimality (agents maximize utility given prices).
- Condition 2: market clearing (feasibility).
- The budget constraint holds with equality since u is strictly increasing.

Solving the Two-agents Problem

- Suppose $u(c) = \log(c)$ $\beta \in (0, 1)$. For an arbitrary agent $i = 1, 2$:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \log(c_t^i) + \lambda_i \left(\sum_{t=0}^{\infty} p_t e_t^i - \sum_{t=0}^{\infty} p_t c_t^i \right) \quad (3)$$

where λ_i is the Lagrange multiplier for the budget constraint of agent i .

- ▶ The solution is interior: $c_t > 0$ for every t ($\lim_{c \rightarrow 0} u'(c) = \infty$).
- ▶ The budget constraint holds with equality (u is strictly increasing).
- FOC: $\frac{\beta^t}{c_t^i} = \lambda_i p_t$ for $t = 0, 1, \dots, \infty$.
- Solving for λ_i in two arbitrary periods:

$$\frac{1}{c_t^i} = \frac{p_t}{p_{t+1}} \frac{\beta}{c_{t+1}^i} \quad \text{for all } t \text{ and } i = 1, 2 \quad (4)$$

Solving the Two-agents Problem

- From the FOC: $c_t^i = c_0^i \frac{p_0}{p_t} \beta^t$. Substituting into the budget constraint ($p_0 = 1$):

$$\sum_{t=0}^{\infty} p_t e_t^i = \sum_{t=0}^{\infty} p_t c_t^i = c_0 \sum_{t=0}^{\infty} \beta^t = \frac{c_0^i}{1 - \beta} \quad (5)$$

- This pins down allocations as a function of prices; now find the equilibrium prices.

Solving the Two-agents Problem

- Equilibrium in the goods market:

$$c_t^1 + c_t^2 = e_t^1 + e_t^2 = \hat{e} \quad \forall t \quad (6)$$

- Summing the FOCs of both agents:

$$c_{t+1}^1 + c_{t+1}^2 = \beta \frac{p_t}{p_{t+1}} (c_t^1 + c_t^2) \quad \forall t \quad (7)$$

- This implies $\hat{e} = \beta \frac{p_t}{p_{t+1}} \hat{e} \Leftrightarrow \beta = \frac{p_{t+1}}{p_t}$. With the normalization $p_0 = 1$:

$$p_t = \beta^t \quad \forall t \quad (8)$$

- Meaning that $c_{t+1}^i = c_t^i = c_0^i$ for both i .

Solving a Dynamic Problem

- Agent 1 receives the endowment in even periods, so:

$$\sum_{t=0}^{\infty} p_t e_t^1 = \hat{e} \sum_{t=0}^{\infty} \beta^{2t} = \frac{\hat{e}}{1 - \beta^2} \quad (9)$$

- For agent 2:

$$\sum_{t=0}^{\infty} p_t e_t^2 = \frac{\hat{e}\beta}{1 - \beta^2} \quad (10)$$

- Equilibrium allocations:

$$c_t^1 = c^1 = \frac{\hat{e}}{1 + \beta} > \frac{\hat{e}}{2} \quad \text{and} \quad c_t^2 = c^2 = \frac{\hat{e}\beta}{1 + \beta} < \frac{\hat{e}}{2} \quad (11)$$

- Agent 1 consumes more because she receives the endowment first.

Sequential Market Structure

- Agents “trade” every period and can borrow or lend at a one-period interest rate r_t .
- Define a_t as the agent’s net position, i.e., savings from period $t - 1$.
- The price of the final consumption good is normalized to 1 *in all periods*.
- The budget constraint for agent i in period t :

$$c_t + a_{t+1} \leq a_t(1 + r_t) + e_t^i. \quad (12)$$

- **Alternative notation (and timing):** bond price $q_t \equiv \frac{1}{1 + r_t}$, so the budget constraint reads $c_t + q_t a_{t+1} = a_t + e_t^i$.

Sequential Markets

Definition 2 (Sequential competitive equilibrium)

A set of sequences $\{c_t^{i*}, a_{t+1}^{i*}\}_{t=0}^{\infty}$ for $i = 1, 2$, and $\{r_t\}_{t=0}^{\infty}$ such that:

1. For each i , $\{c_t^{i*}, a_{t+1}^{i*}\}_{t=0}^{\infty}$ solves

$$\max_{\{c_t^i, a_{t+1}^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t^i) \quad s.t. \quad c_t + a_{t+1} = a_t(1 + r_t) + e_t^i \quad \forall t, a_0^i = 0$$

$$\lim_{T \rightarrow \infty} \frac{a_{T+1}}{\prod_{t=0}^T (1 + r_t)} \geq 0 \quad (\text{no-Ponzi game})$$

2. For all t : $c_t^{1*} + c_t^{2*} = \hat{e}$ and $a_{t+1}^{1*} + a_{t+1}^{2*} = 0$.

Sequential Markets: Equivalence

Proposition (AD $\Leftrightarrow$ Sequential)

Under **complete markets** and the **no-Ponzi game** condition, any Arrow-Debreu equilibrium has an equivalent Sequential Markets equilibrium delivering the same allocations. Prices are related by:

$$p_t = \prod_{s=0}^{t-1} q_s = \frac{1}{\prod_{s=0}^{t-1} (1 + r_s)}.$$

- The AD price p_t is the present value at time 0 of one unit of the good at time t , discounted by the sequence of interest rates.
- What does the **no-Ponzi game** restriction mean?
 - ▶ To give intuition, let's solve the sequential problem in **finite time** and replace NPG with a restriction $a_{T+1} \geq 0$.

Sequential Markets in Finite Horizon

Kuhn-Tucker for agent i :

$$\mathcal{L} = \sum_{t=0}^T [\beta^t u(c_t^i) + \lambda_t (e_t^i + a_t^i(1 + r_t) - c_t^i - a_{t+1}^i)] + \mu_T a_{T+1}^i \quad (13)$$

- Kuhn-Tucker conditions:

- ▶ $a_{T+1} \geq 0$, $\lambda_t \geq 0$, and $\mu_T \geq 0$.
- ▶ Complementary slackness: $a_{T+1}\mu_T = 0$

First-order conditions...

$$\begin{aligned} u'(c_t)\beta^t &= \lambda_t & \text{and} & & \lambda_t &= (1 + r_{t+1})\lambda_{t+1} & \text{for} & & t = 0, \dots, T-1 \\ u'(c_T)\beta^T &= \lambda_T & \text{and} & & \lambda_T &= \mu_T & \text{for} & & t = T \end{aligned}$$

Sequential Markets in Finite Horizon

- Using the FOCs:

$$u'(c_t) = \beta(1 + r_{t+1}) u'(c_{t+1}) \quad t = 0, 1, \dots, T - 1$$

- This is the *Euler Equation* $\Rightarrow$ the most important equation in modern macroeconomics.
 - ▶ Describes the trade-off between consumption and savings.
- In period T :

$$\beta^T u'(c_T) = \lambda_T = \mu_T > 0$$

- ▶ Complementary slackness gives $a_{T+1} = 0 \Rightarrow$ agent doesn't want to die with "money in the pocket".
- ▶ What if we didn't have $a_{T+1} \geq 0$? What does this tell us about the nPg condition?

No-Ponzi Game

- *No-Ponzi game* condition: without it, the agent could always roll over the debt and achieve a higher consumption sequence.
- Substituting the budget constraints up to T (assuming equality):

$$\begin{aligned}\frac{c_0 - e_0}{(1 + r_0)} + \frac{a_1}{(1 + r_0)} &= a_0 \\ \frac{c_1 - e_1}{(1 + r_1)} + \frac{a_2}{(1 + r_1)} &= a_1 \dots \\ \Rightarrow \sum_{t=0}^T \frac{c_t - e_t}{\prod_{j=0}^t (1 + r_j)} + \underbrace{\frac{a_{T+1}}{\prod_{t=0}^T (1 + r_t)}}_{=0 \text{ No-Ponzi-game}} &= a_0\end{aligned}$$

- Alternatively to this condition, we can impose a lower bound such that:

$$a_{t+1} \geq -\bar{A}, \tag{14}$$

provided this lower bound is high enough not to restrict the choice of a_{t+1} .

Sequential Markets in Infinite Horizon

- In infinite time, we don't have the final condition. Using the **Euler Equation** (assuming log):

$$\frac{1}{c_t^i} = (1 + r_{t+1})\beta \frac{1}{c_{t+1}^i} \quad \forall t \text{ and } i = 1, 2.$$

- Note that:

$$\begin{aligned} c_1^i &= (1 + r_1)\beta c_0^i \quad \& \quad c_2^i = (1 + r_2)\beta c_1^i \quad \rightarrow \quad c_2^i = (1 + r_2)(1 + r_1)\beta^2 c_0^i \\ \Rightarrow \quad c_t^i &= c_0^i \beta^t [\prod_{j=1}^t (1 + r_j)] \\ c_t^i &= c_0^i \frac{\beta^t}{1 + r_0} [\prod_{j=0}^t (1 + r_j)] \end{aligned}$$

Sequential Markets in Infinite Horizon

- Substituting $c_t^i = c_0^i \frac{\beta^t}{1 + r_0} [\Pi_{j=0}^t (1 + r_j)]$ into the intertemporal budget constraint:

$$\sum_{t=0}^{\infty} \frac{c_t^i}{\Pi_{j=0}^t (1 + r_j)} + \lim_{T \rightarrow \infty} \frac{a_{T+1}^i}{\Pi_{t=0}^T (1 + r_t)} = a_0^i + \sum_{t=0}^{\infty} \frac{e_t^i}{\Pi_{j=0}^t (1 + r_j)}$$

$$\sum_{t=0}^{\infty} c_0^i \frac{\beta^t}{(1 + r_0)} = a_0^i + \sum_{t=0}^{\infty} \frac{e_t^i}{\Pi_{j=0}^t (1 + r_j)}$$

$$\frac{c_0^i}{(1 - \beta)} = a_0^i (1 + r_0) + \sum_{t=0}^{\infty} \frac{e_t^i}{\Pi_{j=1}^t (1 + r_j)}$$

- Assuming $a_0^i = 0$.
- Without nPg, the intertemporal budget constraint is unbounded.

- Summing the Euler equation of the two agents: $c_t^1 + c_t^2 = (1 + r_{t+1})\beta(c_{t+1}^1 + c_{t+1}^2)$
- Using the goods market equilibrium equation: $c_t^1 + c_t^2 = e_t^1 + e_t^2 = \hat{e}$ for all t :

$$\hat{e} = (1 + r_{t+1})\beta\hat{e} \quad \Rightarrow \quad 1 + r_t = \frac{1}{\beta} \quad \forall t > 0.$$

- ▶ Note that $1/(1 + r_{t+1}) = p_{t+1}/p_t$ from the Arrow-Debreu structure.
- Additionally, $c_t^i = c_0^i \quad \forall t$.

Solution

- Using $\beta = 1/(1 + r_t)$:

$$\frac{c_0^i}{(1 - \beta)} = \sum_{t=0}^{\infty} \frac{e_t^i}{\prod_{j=1}^t (1 + r_j)} = \sum_{t=0}^{\infty} \beta^t e_t^i$$

Substituting the endowment sequences e_t^i , we find the same allocations as in the Arrow-Debreu equilibrium.

- Once we have c_t^i , we can use the budget constraints to calculate savings a_{t+1}^i .
- The equilibrium in the asset market requires: $a_t^1 + a_t^2 = 0$ for all t .

The Social Planner and the Welfare Theorems

The Social Planner

- In the previous section, we solved the model by finding the *decentralized equilibrium*:
 - ▶ **Decentralized equilibrium:** Find the price vector that supports the optimal allocations and CLEARS ALL MARKETS.
- We can also solve for the *optimal* allocations by solving the **social planner's problem**.
- **The “Benevolent” Social Planner’s Problem:**
 - ▶ Maximize the utility of the HHs subject to the technological restrictions and resource constraints (NOT BUDGET CONSTRAINTS).
 - ▶ Does NOT involve any prices.
 - ▶ The solution(s) are the Pareto optimal allocations.

Pareto Optimality

Welfare and Equilibrium

- Under certain conditions, the planner's problem and the decentralized equilibrium yield the same allocations $\Rightarrow$ **Welfare Theorems**.
 - ▶ **First Welfare Theorem:** Competitive Equilibrium $\Rightarrow$ Pareto Optimal. Theorem & Proof
 - ▶ **Second Welfare Theorem:** Pareto Optimal $\Rightarrow$ Competitive Equilibrium. Theorem & Proof
- In this case, the economy is Pareto efficient.
- **When does the First Welfare Theorem not apply?**
 - ▶ Externalities; Incomplete Markets; Imperfect Competition; Asymmetric Information; Distortionary Taxation.

- The Welfare Theorems say that we can go from a Pareto optimal allocation to a decentralized equilibrium and vice versa.
- Under certain conditions, it is sufficient to compute the Pareto optimal allocations by solving the problem of the *Social Planner* (which is generally simpler).
- **Negishi's Method**: Selects the appropriate weight according to the initial endowments of each family to find the allocations of the competitive equilibrium!

Planner's Problem

$$\begin{aligned} \max_{\{c_t^1, c_t^2\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t [\alpha u(c_t^1) + (1 - \alpha)u(c_t^2)] \\ \text{s.t.} \quad & c_t^1 + c_t^2 = e_t^1 + e_t^2 = \hat{e}_t \quad \text{for all } t \\ & c_t^i \geq 0 \quad \text{for all } t \text{ and for all } i \end{aligned}$$

- The $\alpha \in [0, 1]$ defines the relative Pareto weights (e.g., $\alpha = 0.5$ gives equal weight to both agents).
- **Tracing out the Pareto frontier:** As α varies from 0 to 1, the solutions $c_t^i(\alpha)$ trace out *all* Pareto optimal allocations — the entire Pareto frontier.
- The set of Pareto efficient allocations is parameterized by α : different weights give different points on the frontier.

Negishi's Method

- **Key result:** The competitive equilibrium corresponds to a *specific* α^* on the Pareto frontier, determined by initial endowments.
- **Why?** The planner's FOC and the CE Euler equations are equivalent when $\alpha^* = 1/\lambda^1$ (where λ^i is agent i 's budget multiplier) — richer agents get higher weight.
- **Second Welfare Theorem intuition:** Redistributing initial endowments changes λ^i , moving to a different α^* — and that new point is itself a competitive equilibrium.
- **Exercise:** Solve the two-agent problem using Negishi's method.

Equilibrium in the Two-agent Problem

Summary:

- **The decentralized equilibrium:** sequential markets structure vs Arrow-Debreu $\Rightarrow$ “Market” equilibrium.
 - ▶ If markets are complete, the solutions are identical.
- **The benevolent planner’s problem:** Gives the pareto optimal allocations.
- If the welfare theorems are satisfied, the two solutions are identical, and the equilibrium is optimal.

Uncertainty in General Equilibrium

Notation

- An event in period t : $s_t \in S$. S is the set of all possible events, which we assume is finite and equal for all t .
- An event history is a vector represented by: $s^t = (s_0, s_1, \dots, s_t)$.
- Formally $s^t \in S^t$, where $S^t = S \times S \times \dots \times S$.
- The probability of observing a particular history of events is given by: $\pi(s^t)$.
- The conditional probability of observing s^t after the realization of s^τ : $\pi(s^t | s^\tau)$.
- In some places, you may also find the representation of a sub-history of s^t as: $s_{\rightarrow t-1}^t$.

Notation

- Goods are indexed by both period t and the history of events s^t : $c_t(s^t)$.
- An agent chooses a consumption plan dependent on the history of events: $\{c_t(s^t)\}_{t=0}^{\infty}$.
- Agents maximize **expected utility**:

$$U(\{c_t(s^t)\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t \sum_{s^t \in S^t} \pi(s^t) u(c_t(s^t)) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]. \quad (15)$$

- **Example:** Two-agent endowment economy. Agents $i = \{1, 2\}$ receive endowment $e_t^i(s^t)$ depending on history s^t .

Market Structure: Arrow-Debreu

- Trades occur in period 0 before any uncertainty is realized.
- In period 0, agents trade consumption claims in all periods and *possible realizations of s^t* .
- Define the price of a unit of a consumption claim at t and s^t : $p_t(s^t)$.
- The budget constraint of an agent i in period 0:

$$\sum_{t=0}^{\infty} \sum_{s^t \in S^t} p_t(s^t) c_t^i(s^t) \leq \sum_{t=0}^{\infty} \sum_{s^t \in S^t} p_t(s^t) e_t^i(s^t). \quad (16)$$

- *Market clearing* has to be sustained at all dates and possible history of events!

$$c_t^1(s^t) + c_t^2(s^t) = e_t^1(s^t) + e_t^2(s^t) \quad \forall t \text{ and } s^t \in S^t. \quad (17)$$

Market Structure: Arrow-Debreu

Definition 3 (Arrow-Debreu competitive equilibrium with uncertainty)

Sequences $\{c_t^1(s^t), c_t^2(s^t)\}_{t=0, s^t \in S^t}^\infty$ and $\{p_t(s^t)\}_{t=0, s^t \in S^t}^\infty$ such that:

1. Given the sequence of prices $\{p_t(s^t)\}$, for each i , $\{c_t^i(s^t)\}$ solves:

$$\begin{aligned} & \max \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) u(c_t^i(s^t)) \\ \text{s.t.} \quad & \sum_t \sum_{s^t} p_t(s^t) c_t^i(s^t) \leq \sum_t \sum_{s^t} p_t(s^t) e_t^i(s^t). \end{aligned}$$

2. Goods market clears for all t and $s^t \in S^t$:

$$c_t^1(s^t) + c_t^2(s^t) = e_t^1(s^t) + e_t^2(s^t).$$

Market Structure: Arrow-Debreu

- Solution for an arbitrary agent:

$$\mathcal{L} = \sum_{t=0}^{\infty} \sum_{s^t \in S^t} \beta^t \pi(s^t) u(c_t^i(s^t)) + \lambda^i \left(\sum_{t=0}^{\infty} \sum_{s^t \in S^t} p_t(s^t) [e_t^i(s^t) - c_t^i(s^t)] \right) \quad (18)$$

- And the FOCs...

$$\beta^t \pi(s^t) u'(c_t^i(s^t)) = \lambda^i p_t(s^t) \quad \forall t, s^t, i$$

- Note that by substituting with λ , the agent equalizes the marginal utility across different states of nature.

$$\beta^t \frac{\pi(s^t)}{\pi(s_0)} \frac{u'(c_t^i(s^t))}{u'(c_0^i(s_0))} = \frac{p_t(s^t)}{p_0(s_0)} \quad \forall t, s^t, i$$

Market Structure: Arrow-Debreu

- The ratio of marginal utility between agents is constant across all t and s^t :

$$\frac{u'(c_t^2(s^t))}{u'(c_t^1(s^t))} = \frac{u'(c_0^2(s_0))}{u'(c_0^1(s_0))} \quad \forall t, s^t$$

- Example with u CRRA:

$$\left(\frac{c_t^2(s^t)}{c_t^1(s^t)} \right)^{-\sigma} = \left(\frac{c_0^2(s_0)}{c_0^1(s_0)} \right)^{-\sigma} \quad \forall t, s^t$$

$\Rightarrow$ The consumption ratio between two agents is constant across all t and s^t .

- Given the resource constraint: $c_t^1(s^t) + c_t^2(s^t) = e_t^1(s^t) + e_t^2(s^t) = e_t(s^t)$: an agent consumes a constant fraction θ^i of the aggregate endowment $e_t(s^t)$.

Market Structure: Arrow-Debreu

- There is **perfect risk sharing**: each agent consumes a constant fraction θ^i of aggregate income — individual consumption fluctuates only with aggregate endowment, not with the idiosyncratic distribution.
- All trades are negotiated at $t = 0$, so the consumption *share* θ^i is determined once and for all by initial wealth.
- This requires perfect information and enforceable contracts (*full commitment*).

Market Structure: Arrow-Debreu

- Solving for prices, we use optimality + resource constraint:

$$\begin{aligned} p_t(s^t) &= \beta^t \frac{\pi(s^t)}{\pi(s_0)} \left(\frac{c_t^i(s^t)}{c_0^i(s_0)} \right)^{-\sigma} \\ &= \beta^t \frac{\pi(s^t)}{\pi(s_0)} \left(\frac{e_t(s^t)}{e_0(s_0)} \right)^{-\sigma} \end{aligned}$$

- That is, the “price” of consumption in a state of nature s^t depends on the probability that this state is realized and the amount of aggregate wealth ($e_t(s^t)$).
- The price is high in “lean times” (low $e_t(s^t)$): everyone wants insurance precisely when resources are scarce.

Market Structure: Sequential Markets

- Let's define a sequential market structure. In every period, markets open, and trades take place.
- For the equivalence between Arrow-Debreu and sequential markets with uncertainty, we need to deliver one unit of consumption in **all states of nature**.
- The agent can buy a contract at the price of $q_t(s_{t+1}, s^t)$ in period t and history s^t , which delivers one unit of consumption in the next period and state s_{t+1} , for each event s_{t+1} .
- The agent can, in period t , fully protect against any event that will occur in $t + 1$ by buying a contract for each s_{t+1} .
- These financial instruments are known as: **Arrow securities**.
- In the case where it is possible to trade *Arrow securities* in all periods and states of nature, Arrow (1964) shows that we can trade *goods* between different t and s^t , that is, we have **complete markets**.

Market Structure: Sequential Markets

- Define $a_{t+1}(s_{t+1}, s^t)$ as the quantity of Arrow securities bought by agents in period t .
- The budget constraint of an arbitrary agent i in t and s^t :

$$c_t^i(s^t) + \sum_{s_{t+1}} a_{t+1}^i(s_{t+1}, s^t) q(s_{t+1}, s^t) \leq e_t^i(s^t) + a_t^i(s^t)$$

- Note that agents buy Arrow securities in t for all contingencies $s_{t+1} \in S$, but once s_{t+1} is realized, the position of $t + 1$ is only $a_{t+1}(s_{t+1}, s^t)$ corresponding to the realized state.
- The Arrow securities market needs to clear at zero for all periods and events.

Market Structure: Sequential Markets

Definition 4 (Sequential competitive equilibrium with uncertainty)

Sequences $\{c_t^i(s^t), a_{t+1}^i(s_{t+1}, s^t)\}$ and prices $\{q_t(s_{t+1}, s^t)\}$ such that:

1. Given the sequence of prices $\{q_t(s_{t+1}, s^t)\}$, for each i , $\{c_t^i, a_{t+1}^i\}$ solves:

$$\begin{aligned} & \max_{t, s^t} \sum \beta^t \pi(s^t) u(c_t^i(s^t)) \\ \text{s.t. } & c_t^i(s^t) + \sum_{s_{t+1}} a_{t+1}^i(s_{t+1}, s^t) q_t(s_{t+1}, s^t) \leq e_t^i(s^t) + a_t^i(s^t), \quad \forall t, s^t \\ & a_{t+1}^i \geq -\bar{A}^i, \quad a_0^i \text{ given.} \end{aligned}$$

2. Goods and asset markets clear for all t, s^t, s_{t+1} :

$$\begin{aligned} c_t^1(s^t) + c_t^2(s^t) &= e_t^1(s^t) + e_t^2(s^t), \\ a_{t+1}^1(s_{t+1}, s^t) + a_{t+1}^2(s_{t+1}, s^t) &= 0. \end{aligned}$$

Market Structure: Sequential Markets

- Solution for an arbitrary agent:

$$\mathcal{L} = \sum_{t=0}^{\infty} \left(\sum_{s^t \in S^t} \beta^t \pi(s^t) u(c_t^i(s^t)) + \dots \right. \\ \left. \dots \sum_{s^t \in S^t} \lambda_t^i(s^t) \left[e_t^i(s^t) + a_t^i(s^t) - c_t^i(s^t) - \sum_{s_{t+1}} a_{t+1}^i(s_{t+1}, s^t) q_t(s_{t+1}, s^t) \right] \right)$$

- The optimality conditions, where $\lambda_{t+1}^i(s_{t+1}, s^t)$ is the multiplier for s_{t+1} given a history s^t :

$$\beta^t \pi(s^t) u'(c_t^i(s^t)) = \lambda_t^i(s^t) \quad \forall t, s^t, i$$

$$\lambda_t^i(s^t) q_t(s_{t+1}, s^t) = \lambda_{t+1}^i(s_{t+1}, s^t)$$

- Note the equivalence between Arrow-Debreu and sequential when:

$$q_t(s_{t+1}, s^t) = \frac{p_{t+1}(s^{t+1})}{p_t(s^t)}$$

Market Structure: Sequential Markets

- In other words, the *pricing kernel* is:

$$q_t(s_{t+1}, s^t) = \beta \frac{u'(c_{t+1}^i(s^{t+1}))}{u'(c_t^i(s^t))} \pi(s^{t+1}|s^t)$$

where $\pi(s^{t+1}|s^t) = \pi(s_{t+1}, s^t)/\pi(s^t)$.

- The price of **one** Arrow security associated with the state s_{t+1} . Remember that Arrow securities only pay off in one state of nature (in the others, they pay 0).
- The *pricing kernel* is widely used in macro-finance, and from it, we can price various assets.

Market Structure: Sequential Markets

$$\underbrace{q_t(s_{t+1}, s^t)}_{\text{price of the security that pays in state } s_{t+1}} = \underbrace{\beta \frac{u'(c_{t+1}^i(s^{t+1}))}{u'(c_t^i(s^t))}}_{\text{ratio of mg. util. received in state } s_{t+1}} \underbrace{\pi(s^{t+1}|s^t)}_{\text{prob. that state } s_{t+1} \text{ happens}}$$

- Higher probability increases the price of the security.
- Higher mg. utility in s_{t+1} increases the price of the security.

Pricing an Asset

- It is useful to price assets in terms of real returns. Define the one-period realized real return of an asset j between s^t and s^{t+1} :

$$R_{t+1}^j(s^{t+1}) = \frac{P_{t+1}^j(s^{t+1}) + d_{t+1}^j(s^{t+1})}{P_t^j(s^t)}$$

where $P_t^j(s^t)$ and $d_t^j(s^t)$ is the price of the asset j , in time t and state s^t .

- An arrow security (pays dividend = 1 in state s^{t+1} and nothing else in other states nor the future), has gross returns of:

$$R_{t+1}^A(s^{t+1}) = \frac{0 + 1}{q(s_{t+1}, s^t)} = \frac{1}{q(s_{t+1}, s^t)}$$

Example: Price of Risk-free Bond

- What is the price of a risk-free bond that always pay 1 in the next period (and nothing afterwise)?

$$R_{t+1} = \frac{0 + 1}{P_t^{\text{risk free}}(s^t)}$$

- The price of the risk-free bond equals the sum of all Arrow security prices:

$$P_t^{\text{risk free}}(s^t) = \sum_{s_{t+1}|s^t} q_t(s_{t+1}, s^t)$$

Example: Price of Risk-free Bond

- Thus, the return of a risk-free bond (non-contingent on the state):

$$\sum_{s_{t+1}|s^t} q_t(s_{t+1}, s^t) = R_{t+1}^{-1}.$$

- In other words, the price of **full consumption insurance** in t is the sum of the prices of Arrow securities associated with all events s_{t+1} :

$$\sum_{s_{t+1}|s^t} q_t(s_{t+1}, s^t) = \beta \sum_{s_{t+1}|s^t} \frac{u'(c_{t+1}^i(s^{t+1}))}{u'(c_t^i(s^t))} \pi(s^{t+1}|s^t)$$

Example: Price of Risk-free Bond

- Note that $\sum_{s^{t+1}} \pi(s^{t+1}|s^t) u'(c_{t+1}^i(s^{t+1})) = \mathbb{E}_t [u'(c_{t+1}^i)]$ is the conditional expected marginal utility of consumption given information in t .
- Substituting R_{t+1}^{-1} and the conditional expectation, we can rewrite the Euler Equation:

$$u'(c_t^i(s^t)) = \beta R_{t+1} \mathbb{E}_t [u'(c_{t+1}^i(s^{t+1}))] \quad \forall t, s^t.$$

- This is the Euler equation you will encounter most of the time.

Taking Stock

- How to solve a **Dynamic General Equilibrium** model?
 1. Describe the economy's environment;
 2. Solve the agents' problem;
 3. Specify the equilibrium conditions;
 4. Describe the competitive equilibrium.
- How to use the **Welfare Theorems** to solve the model?
 - ▶ Given certain conditions, the solution of the **Central Planner** is equivalent to the decentralized equilibrium.
 - ▶ In this case, we also know that the equilibrium is Pareto efficient.
- We also have seen that if we have Arrow securities available in all periods and state of nature, markets are complete and the solution of an Arrow-Debreu structure is equivalent to the sequential market.

Appendix

- Suppose an arbitrary economy: N goods, H households (utility U^h , endowments e^h), F firms (output y_j^f), ownership θ_h^f with $\sum_h \theta_h^f = 1$.
- **Definition (Feasibility):** An allocation $\{x_j^h, y_j^f\}$ is feasible if for every j :

$$\sum_h x_j^h \leq \sum_h e_j^h + \sum_f y_j^f \quad (19)$$

- **Definition (Pareto Optimality):** A feasible allocation $\{x_j^h, y_j^f\}$ is Pareto optimal if there is no other feasible allocation $\{\hat{x}_j^h, \hat{y}_j^f\}$ such that

$$U^h(\{\hat{x}_j^h\}) \geq U^h(\{x_j^h\}) \quad \text{for every } h \quad (20)$$

$$U^h(\{\hat{x}_j^h\}) > U^h(\{x_j^h\}) \quad \text{for at least one } h. \quad (21)$$

First Welfare Theorem

← Back

Theorem (First Welfare Theorem)

Suppose that $\{x_j^h, y_j^f, p_j\}$ is a competitive equilibrium, and all U^h are locally nonsatiated. Then $\{x_j^h, y_j^f\}$ is Pareto optimal.

- **Proof:** By contradiction. Suppose $\{x_j^h, y_j^f\}$ is not Pareto optimal (i.e., there exists another feasible allocation that gives more utility to at least one h) and use the definition of a competitive equilibrium.
- Note that we are assuming the existence of a competitive equilibrium (which may not exist depending on the form of U^h , and the sets of x and y).
- Pareto optimality says nothing about equity (an individual consuming everything is efficient).

Second Welfare Theorem

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Theorem (Second Welfare Theorem)

Consider the Pareto optimal allocation $\{x_j^h, y_j^f\}$. Given certain conditions (convex production and consumption sets, utility concave, continuous, and locally nonsatiated), there exists a competitive equilibrium with prices $\{p_j\}$ and endowments $\{e^h, \theta_h^f\}$ that supports the allocation $\{x_j^h, y_j^f\}$.

- **Proof:** More involved — requires showing the existence of prices (on a hyperplane) that support the allocations.
- Given an appropriate redistribution of initial endowments, we can decentralize any Pareto optimal allocation as a competitive equilibrium.